

Mark schemes

Q1.

(a) $h = 2$

PI

B1

$$g(x) = \sqrt{x^3 + 1}$$

$$I \approx h/2 \{...\}$$

$$\{...\} = g(0) + g(6) + 2[g(2) + g(4)]$$

OE summing of areas of the 'trapezia'.

*Can award even if MR expression for $g(x)$
but must be using from 0 to 6*

M1

$$\begin{aligned} \{...\} &= 1 + \sqrt{217} + 2(3 + \sqrt{65}) \\ &= 1 + 14.73... + 2(3 + 8.06...) \end{aligned}$$

OE Accept 2 dp evidence for surds

A1

$$(I \approx) 37.8554... = 37.86 \text{ (to 4 sf)}$$

Must be 37.86

A1

4

(b) $f(x) = \sqrt{(2x)^3 + 1} = \sqrt{8x^3 + 1}$

$$\sqrt{kx^3 + 1}, \quad k \neq 1 \text{ or } 0 \text{ or } f(x) = g(2x) \quad \text{All rights reserved.}$$

M1

Either form acceptable

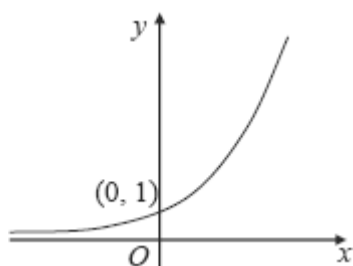
A1

2

[6]

Q2.

(a)



Shape (graph must clearly go below the intersection pt.). Condone if x -axis is a tangent

B1

Only intersection with y -axis at $(0, 1)$ stated/indicated ... (accept 1 on y -axis as equivalent) 0

B1

2

- (b) (i) Stretch (I) in x -direction (II) scale factor 0.5 (III)
Need (I) & one of (II), (III)

M0 if >1 transformation

M1

A1

2

- (ii) Translation;
Must be 'Translation' or 'translate (d)' for 1st B mark

B1;

$$\begin{bmatrix} -1 \\ 0 \end{bmatrix}$$

*Accept **full** equivalent to vector in words provided linked to 'translation/move/shift' and **negative** x -direction (Note: B0 B1 is possible)*

B1

ALTn: Stretch (I) in y -direction (II) scale factor 3 (III)
[Mark the alternative as in (b)(i).]

2

- (c) (i) $9^x = (3^2)^x = 3^{2x} = (3^x)^2 = Y^2$;
Justifying either $9^x = Y^2$ or $3^{x+1} = 3Y$

M1

$$\begin{aligned}
 3^{x+1} &= 3^x \times 3^1 = 3Y \\
 9^x - 3^{x+1} + 2 &= 0 \Rightarrow Y^2 - 3Y + 2 = 0 \\
 &\Rightarrow (Y-1)(Y-2) = 0 \\
 &\text{AG}
 \end{aligned}$$

A1

2

(ii) $Y = 1 \Rightarrow 3^x = 1 \Rightarrow x = 0$
 AG (Accept direct substitution if convinced)

B1

$$\begin{aligned}
 Y = 2 &\Rightarrow 3^x = 2 \\
 \log_{10} 3^x &= \log_{10} 2 \\
 &\text{Takes logs of both, PI by 'correct' value(s) later.} \\
 &\text{or } x = \log_3 2 \text{ seen}
 \end{aligned}$$

M1

$$\begin{aligned}
 x \log_{10} 3 &= \log_{10} 2 \\
 &\text{Use of } \log 3^x = x \log 3 \text{ or} \\
 &\frac{\lg 2}{\lg 3} \\
 \log_3 2 &= \frac{\lg 2}{\lg 3} \text{ OE (PI by } \log_3 2 = 0.630 \text{ or} \\
 &0.631 \text{ or better)}
 \end{aligned}$$

m1

$$x = \frac{\lg 2}{\lg 3} = 0.630929 \dots = 0.6309 \text{ to 4 dp}$$

Must show that logarithms have been used
 otherwise 0/3

A1

4

[12]

Q3.

Stretch (I)

Scale factor $\frac{1}{2}$ (II)
 $I + (II \text{ or } III)$

M1

parallel to x -axis (III)
 All correct

A1

(Or scale factor 4 parallel to y -axis)
Translation

M1

$$\begin{bmatrix} 0 \\ -5 \end{bmatrix} \quad \text{OE}$$

A1

Alternatives:

translate $\begin{pmatrix} 0 \\ -\frac{5}{4} \end{pmatrix}$, stretch sf 4 $\parallel y$ -axis
translate $\begin{pmatrix} 0 \\ -5 \end{pmatrix}$, stretch sf $\frac{1}{2}$ $\parallel x$ -axis

Mark translation first. Mark stretch as above, but relative to their translation.

[4]

Q4.

(a) $\left(x - \frac{3}{2}\right)^2$

Must have $()^2$ $p = 1.5$

B1

$$+ \frac{7}{4}$$

$$q = 1.75$$

B1

2

(b) Minimum value is $\frac{7}{4}$

ft their q or correct value

B1ft

1

(c) **Translation** (and **no other** transformation stated)
(not shift, move, transformation etc)

E1

through $\begin{bmatrix} 3 \\ 2 \\ 7 \\ 4 \end{bmatrix}$ (or equivalent in words)
M1 for one component correct
or ft their p or q values

M1

CSO; condone 1.5 right and 1.75 up etc

A1

3

[6]

Q5.

- (a) (i) $h = 0.5$
 Integral = $h/2 \{ \dots \}$
PI

B1

$$\{ \dots \} = f(0) + 2\left[f\left(\frac{1}{2}\right) + f(1) + f\left(\frac{3}{2}\right)\right] + f(2)$$

OE summing of areas of the four traps.

M1

$$\{ \dots \} = 1 + 2[\sqrt{6} + 6 + 6\sqrt{6}] + 36$$

$$= 1 + 2[2.449.. + 6 + 14.6969..] + 36$$

Condone 1 numerical slip. Accept 3sf values if not exact.

A1

$$= 37 + 2 \times 23.146.. = 83.292...$$

$$\text{Integral} = 0.25 \times 83.292.. = 20.8 \text{ (3sf)}$$

CAO; must be 20.8

A1

4

- (ii) Relevant trapezia drawn on a copy of given graph
Accept single trapezium with its sloping side above the curve

M1

{Approximation is an} overestimate
Dep. on 4 trapezia with each of their

upper vertices lying on the curve

A1

2

- (b) (i) Stretch (I) in x -direction (II)
Need (I) and one of (II), (III)
M0 if more than one transformation

M1

(scale factor) $\frac{1}{3}$ (III)

A1

2

- (ii) $6^{3x} = 84$
PI

M1

$$\log_{10} 6^{3x} = \log_{10} 84$$

*Take logs of both sides of $a^x = b$, PI by
'correct' value(s) later or $3x = \log_6 84$*

M1

$$3x \log_{10} 6 = \log_{10} 84$$

*Use of $3 \log 6^{3x} = 3x \log 6$ OE
or $3x = \log_6 84$ seen*

m1

$$x = \frac{\lg 84}{3 \lg 6}$$

$$x = 0.82429\dots = 0.824 \text{ (to 3dp)}$$

*Must see that logs have been used before
any of the last 3 marks are awarded in
(b)(ii). Condone > 3dp*

A1

4

- (c) $f(x) = 6^{x-1} - 2$
B1 For either $6^{x-1} + 2$ or for $6^{x+1} - 2$

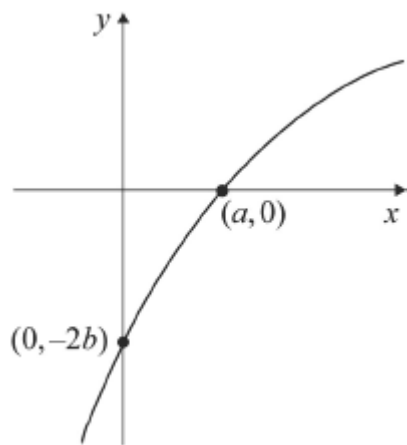
B2,1

2

[14]

Q6.

(a)



shape

coordinates

B1

B1

2

(b) (i) Translation

$$\begin{bmatrix} -1 \\ 0 \end{bmatrix}$$

M1

A1

Stretch I
I + (II or III)

SF 4 II
// y-axis III
I + II + III

M1

A1

Translation $\begin{bmatrix} 0 \\ -2 \end{bmatrix}$
both

B1

All correct and no mistakes on order etc

A1

OR

I stretch M1 I + (II or III)

II SF 4

III // y-axis A1 (I + II + III)

Translation M1

$$\begin{bmatrix} -1 \\ -2 \end{bmatrix} \quad \begin{matrix} A1 \\ B1 \end{matrix}$$

All correct A1

6

Alternative:

$$y = 4 \ln(x + 1) - 2 = 4 \left[\ln(x + 1) - \frac{1}{2} \right]$$

(B1)

Translation

(M1)

$$\begin{bmatrix} -1 \\ -\frac{1}{2} \end{bmatrix}$$

(A1)

Stretch I

I + (II or III)

(M1)

SF 4 II

// y-axis III

I + II + III

(A1)

All correct and no mistakes on order etc

(A1)

(6)

(ii) $y = 4 \ln(x + 1) - 2$
 $x = 0 \quad y = -2$

B1

$$y = 0$$

$$4 \ln(x + 1) = 2$$

$$\ln(x+1) = \frac{1}{2}$$

isolate $\ln(x+1) =$ or $(x+1)^4$

M1

$$x+1 = e^{\frac{1}{2}}$$

$$x+1 = e^k$$

A1

$$x = e^{\frac{1}{2}} - 1 \quad \text{o.e.}$$

CSO isw

A1

4

[12]

Q7.

(a) $\{x =\} \cos^{-1}(0.3) = 1.266.... \{ = \beta \}$
 $\cos^{-1}(0.3)$ PI by eg 72° or 73°

M1

$$\{x =\} 2\pi - \beta$$

Condone degrees or mix.

m1

$$x = 1.27, 5.02$$

Accept 1.26 to 1.27 with 5.01 to 5.02 inclusive

A1

3

(b) (i) $m(\pi, -1)$
B1 for each coordinate

B1;B1

2

(ii) $\{x_Q =\} = 2\pi - \alpha$
OE (unsimplified)

B1

1

(c) Stretch (I) in x -direction (II) scale
Need(I) & one of (II), (III)

M1

factor $\frac{1}{2}$ (III)

A1

2

(d) $\cos 2x = \cos \frac{4\pi}{5} \Rightarrow 2x = \frac{4\pi}{5}$

OE. (From correct work)

B1

$\Rightarrow x = \frac{2\pi}{5} (= \alpha)$

Condone decimals/degrees

$x = \pi - \alpha$; OE

OE eg $2x = 2\pi - \frac{4\pi}{5}$

Correct quadrant;
condone degrees/decimals/mix

M1

$x = \pi + \alpha$; $x = 2\pi - \alpha$; OE

Need both (OE for $2x =$) with no extras
(quadrants) within the given interval.
Condone degrees/decimals/mix

m1

$x = \frac{2\pi}{5}, \frac{3\pi}{5}, \frac{7\pi}{5}, \frac{8\pi}{5}$

Need all 4 solutions for x but condone
unsimplified provided in terms of π
Ignore extra values outside the given interval.

A1

4

[12]

Q8.

Stretch (I)

SF $\frac{1}{3}$ (II)

For I + (II or III)

M1

Parallel to x -axis (III)

All correct

A1

Translate

Allow translation

E1

$$\begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

Correct vector or description

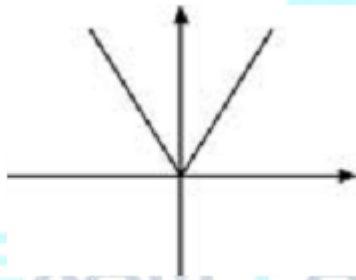
B1

4

[4]

Q9.

(a)

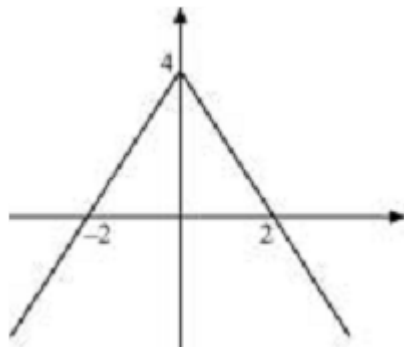


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B1

1

(b)



Shape inverted V in all four quadrants

M1

Symmetrical about y-axis

A1

Coordinates

A1

3

(c) $4 - |2x| = x$

M1

$$4 - 2x = x \quad x = \frac{4}{3}$$

Attempt to solve

A1

$$4 + 2x = x \quad x = -4$$

And no others

A1

3

(d) $-4 < x < \frac{4}{3}$

Either correct

M1

Other solution and no extras

$$SC - 4 \leq x \leq \frac{4}{3} \quad B1$$

A1

2

[9]

Q10.

(a) (i) $(x + 5)^2$

$$p = 5$$

B1

$$-6$$

$$q = -6$$

B1

2

- (ii) $x_{\text{vertex}} = -5$ (or their $-p$)
may differentiate but must have $x = -5$

B1 ✓

$y_{\text{vertex}} = -6$ (or their q)
and $y = -6$. Vertex $(-5, -6)$

B1 ✓

2

- (iii) $x = -5$

B1

1

- (iv) Translation (not shift, move etc)
and NO other transformation stated

E1

through $\begin{bmatrix} -5 \\ -6 \end{bmatrix}$ (or 5 left, 6 down)
either component correct

M1

M1, A1 independent of E mark

A1

3

- (b) $x+11 = x^2 + 10x + 19$
quadratic with all terms on one side of equation

$\Rightarrow x^2 + 9x + 8 = 0$ or $y^2 - 13y + 30 = 0$

M1

$(x+8)(x+1) = 0$ or $(y-3)(y-10) = 0$

attempt at formula (1 slip) or to factorise

ml

$\left. \begin{matrix} x = -1 \\ y = 10 \end{matrix} \right\}$ or $\left. \begin{matrix} x = -8 \\ y = 3 \end{matrix} \right\}$

both x values correct

A1

both y values correct and linked
SC $(-1, 10)$ **B2**, $(-8, 3)$ **B2** no working

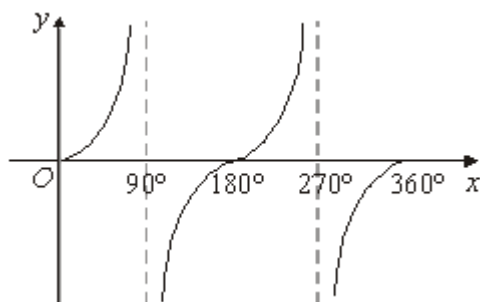
A1

4

[12]

Q11.

(a)



Correct shape of branch from O to 90°
 or correct shapes of branches from 90° - 360°

M1

Complete graph for $0^\circ \leq x \leq 360^\circ$
 (Asymptotes not explicitly required but graphs should show 'tendency')

A1

Correct scaling on x-axis $0^\circ \leq x \leq 360^\circ$

A1

3

(b) 61° ;

For 61°

B1

241°

For 241° and no 'extras' in the interval $0^\circ \leq x \leq 360^\circ$

B1

2

(c) (i) $\sin \theta = -\cos \theta \Rightarrow \frac{\sin \theta}{\cos \theta} = -1$
 $\Rightarrow \tan \theta = -1$

AG; be convinced that the identity $\frac{\sin \theta}{\cos \theta} = \tan \theta$
 is known and validly used

B1

1

(ii) $\Rightarrow \tan(x - 20^\circ) = -1$

M1

$$x - 20^\circ = \tan^{-1}(-1)$$

m1

$$x - 20^\circ = 135^\circ, 315^\circ \dots$$

$$x = 155^\circ;$$

A1

$$335^\circ$$

Ft on (180 + "155") and no 'extras' in the given interval.

A1ft

4

(d) Translation

'Translation'/'translate(d)'

B1

$$\begin{bmatrix} 20 \\ 0 \end{bmatrix}$$

Accept equivalent in words provided linked to 'translation/move/shift'

(Note: B0B1 is possible)

B1

2

(e) $f(x) = \tan 4x$

For $\tan 4x$

B1

1

[13]

Q12.

(a) (i) Stretch (I) in y -direction (II)

Scale factor 2 (III)

>1 transformation is M0.

M1 for (I) and either (II) or (III) or (III)

M1A1

2

(ii) Reflection;

M1

in x -axis

*'Reflection'/'reflect(ed)'
(or in y -axis or $y = 0$ or $x = 0$)*

A1

2

(iii) Translation;

'Translation'/'translate (d)'

B1

$$\begin{bmatrix} 30^\circ \\ 0 \end{bmatrix}$$

*Accept **full** equivalent in words provided linked
to 'translation/move/shift' and **positive** x -direction
(Note: B0 B1 is possible)*

B1

2

(b) $\{\theta - 30^\circ = \sin^{-1}(0.7) = 44.4\dots^\circ$

Inverse sine of 0.7 PI eg by sight of 44, 74 or better

M1

$\dots\dots\dots = 180^\circ - 44.4^\circ$

Valid method for 2nd angle

m1

$\theta = 74.4^\circ, 165.6^\circ$

Condone >1dp accuracy

A1

3

(c) $\dots = \cos^2 x + 2\cos x \sin x + \sin^2 x + \cos^2 x - 2\cos x \sin x + \sin^2 x$

Award for either bracket expanded correctly

M1

$\dots = 2\cos^2 x + 2\sin^2 x$

OE

A1

$$= 2(\cos^2 x + \sin^2 x) = 2 \quad (1)$$

$\cos^2 x + \sin^2 x = 1$ stated or used.

M1

$$= 2$$

AG (be convinced)

A1

4

[13]

Q13.

(a) $(x + 4)^2$

$$p = 4$$

B1

$$+3$$

$$q = 3$$

B1

2

(b) $(x + 4)^2 = -3$ or "their" $(x + p)^2 = -q$
 Or discriminant = $64 - 76$

M1

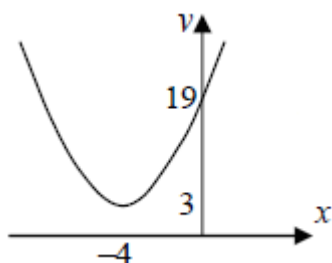
No real square root of -3

Disc < 0 so no real roots (all correct figs)

A1

2

(c)



Minimum $(-4, 3)$

ft their $-p$ and q (or correct)

B1 ✓

graph

Parabola (vertex roughly as shown)

B1

Crossing at $y = 19$ marked or $(0, 19)$ stated

B1

3

(d) Translation (and no additional transformation)

Not shift, move, transformation, etc

E1

through $\begin{bmatrix} -4 \\ 3 \end{bmatrix}$

One component correct eg 3 units up

M1

All correct – if not vector – must say 4 units in negative x -direction, to left etc

A1

3

[10]

Q14.

(a) (i) y -coordinate of A is $27 - 3^0 = 26$

M1A1

2

(ii) When $x = 3$, $y = 27 - 3^3 = 0 \Rightarrow B(3,0)$

AG; be convinced

B1

1

(b) $h = 1$

PI

B1

$\text{Area} \approx h/2\{\dots\}$

$\{\dots\} = f(0) + f(3) + 2[f(1) + f(2)]$

OE summing of areas of the 'trapezia'..

M1

$\{\dots\} = "26" + 0 + 2(24 + 18)$

on (a)(i) ($\Sigma \text{trap} = "25" + 21 + 9$)

A1 ✓

(Area $\approx$) 55

on $[42 + 0.5 \times "(a)(i)"]$

A1 ✓

4

(c) (i) $\log_{10} 3^x = \log_{10} 13$

Takes ln or $\log_{10}$ on both or $x = \log_3 13$

M1

$$x \log_{10} 3 = \log_{10} 13$$

*Use of $\log 3^x = x \log 3$ or $\log_3 13 = \frac{\lg 13}{\lg 3}$ OE
(PI by $\log_3 13 = 2.335$ or better)*

m1

$$x = \frac{\lg 13}{\lg 3} = 2.334717$$

.... = 2.3347 to 4dp

Must show that logarithms have been used

A1

3

(ii) $\{k =\} 14$

Condone $y = 14$; Accept final answer 14 with only zeros after decimal point e.g. 14.000

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B1

1

(d) (i) Translation;

'Translation' / 'translate(d)'

B0 if more than one transformation

B1

$$\begin{bmatrix} 0 \\ -27 \end{bmatrix}$$

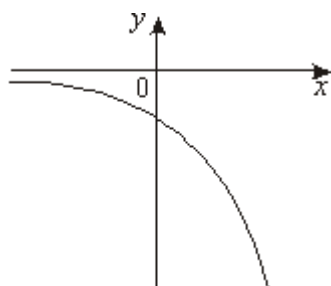
Accept full equivalent in words provided

linked to 'translation/move/shift' and negative y-direction

(Note: B0 B1 is possible)

B1

(ii)



Correct shape (translation of given curve vertically downwards)

B1

Only point of intersection with coordinate axes is on negative y -axis and curve is asymptotic to the negative x -axis

B1

2

[15]

Q15.

- (a) Stretch (I) in x -direction (II) scale
Need(I) and one of (II),(III)

M1

factor 2 (III)

M0 if more than one transformation

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A1

2

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- (b) $\tan^{-1} 3 = 1.2(49....) (= \alpha)$
 $\tan^{-1} 3$ [PI by 71.(56..)]

M1

$$\left\{ \frac{1}{2}x \right\} = \pi + \alpha ;$$

Correct quadrant; condone degrees or mix

m1

$$\frac{1}{2}x = 1.249...; 4.3906...$$

$$x = 2.498... = 2.50 \text{ to 3 sf}$$

A1

$$x = 8.781... = 8.78 \text{ to 3 sf}$$

A1

*Condone 2.5 otherwise deduct **max** of 1 mark throughout this question from A marks if 'correct' rads. but to 2sf or final answers in degrees. (143°, 503°)*
As usual, accept greater accuracy answers.
Ignore extra values outside the given interval (0 to 12.6). If > 2 values inside interval lose an A mark for each one.
NB M1m0A1A0 is possible

SC after M0 for error $\tan x = 6$;

Either $x = 1.40(5)$, $4.54(7)$, $7.68(8)$, $10.8(3)$ or $x = 80.5^\circ$, 260.5° , 440.5° , 620.5° SC
B1

(accept each rounded or truncated to 3 sf)

4

(c) $\cos \theta = 0$, $\sin \theta - 3 \cos \theta = 0$

Need both

M1

$$\tan \theta = \frac{\sin \theta}{\cos \theta} \text{ or } \tan \theta = 3$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta} \text{ seen/used}$$

M1

$$\cos \theta = 0 \Rightarrow \theta = \frac{\pi}{2} = 1.57(07...)$$

Accept $\frac{\pi}{2}$

B1

or $\theta = \frac{3\pi}{2} = 4.71(23..)$

Accept $\frac{3\pi}{2}$

B1

$$\tan \theta = 3 \Rightarrow \theta = 1.249...; 4.3906... = 1.25, 4.39 \text{ to 3sf}$$

If not correct, ft on (b)

NB M0M1(B0B0)A1ft is possible

A1 ✓

$90^\circ; 270^\circ; 71.5(6)^\circ; 251.5(6)^\circ$

5

[11]

Q16.

(a) $A(0^\circ, 1)$

Condone radians

B1

$B(45^\circ, 0)$

Condone (0.785,0) or better.

B1

$C(270^\circ, -1)$

B1 for 270; B1 for -1

B1, B1

4

(b) Stretch (I) in x -direction (II) with a scale factor $\frac{1}{2}$ (III)

More than one transformation is M0

M1 for (I) and either (II) or (III)

M1A1

2

(c) $\cos^{-1}0.37 = "68.284..." (= \alpha)$

$\cos^{-1}0.37$ (PI eg by 68.3 or 1.19)

M1

$$x = \frac{\alpha}{2} = 34.1(42.)^\circ$$

Condone 34.2° , 34° or 0.596 rads

A1

$$x = 180 - \frac{\alpha}{2}$$

OE eg $2x = 360 - \alpha$

m1

$$x = 180 + \frac{\alpha}{2} \quad \text{and} \quad x = 180 + 180 - \frac{\alpha}{2}$$

*OE Need both (OE for $2x =$)
 with no extras (quadrants)
 within the given interval*

m1

$$2x = 38.284\dots; 291.175\dots;$$

$$428.284\dots; 651.715\dots$$

$$x = (34.1^\circ;)$$

$$145.9^\circ; 214.1^\circ; 325.9^\circ$$

*Dep. on **all** three method marks. Must be in degrees*

A1

5

[11]

Q17.

- (a) {y-coordinate of A is} 2

B1

1

- (b) (i) $h = 0.25$

B1

$$\text{Integral} = \frac{h}{2} \{ \dots \}$$

$$\{ \dots \} =$$

$$f(0) + 2\left[f\left(\frac{1}{4}\right) + f\left(\frac{1}{2}\right) + f\left(\frac{3}{4}\right)\right] + f(1)$$

$$\{ \dots \} =$$

M1

$$2 + 4 + 2[(2.316\dots + 2.732\dots + 3.279(5.))]$$

A1 ✓

$$\{ = 6 + 2 \times 8.3276\dots \} \{ = 22.65(5\dots) \}$$

*Condone one numerical slip
 Accept values to 3sf (rnd or trunc)
 ft answer from (a) if not "2"*

$$\text{Integral} = 0.125 \times 22.655\dots = 2.8319\dots$$

$$\text{Integral} = 2.83 \text{ to 3 sf}$$

CAO Must be 2.83
 (NMS scores 0/4)

A1

4

- (ii) Relevant trapezia drawn on a copy of given graph
Accept relevant single trapezium with its sloping side above the curve

M1

{Approximation is an}overestimate

A1

2

(c) $5 = 3^x + 1 \Rightarrow 3^x = 4$

B1

$$\log_{10} 3^x = \log_{10} 4$$

Takes ln or $\log_{10}$ on both sides of $3^x = k$, where $k > 0$

M1

$$x \log_{10} 3 = \log_{10} 4$$

Use of $\log 3^x = x \log 3$

m1

$$x = \frac{\lg 4}{\lg 3} = 1.26185 \dots = 1.2619 \text{ to 4dp}$$

*Accept 4dp or better
 [If using T&I a full justification is required; else M0m0A0]*

A1

4

(d) $f(x) = 3^{-x} + 1$

B1

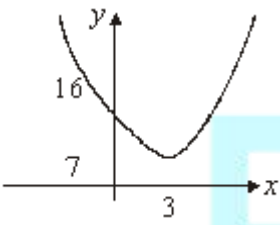
1

[12]

Q18.

(a) $(x - 3)^2$

$$p = 3$$

			B1	
	+ 7			
	$q = 7$			
			B1	2
(b)	(i)	Vertex (3, 7) <i>ft their p</i>		
		<i>ft their q</i>	B1ft	
			B1ft	2
	(ii)	graph		
		 <i>parabola (ft on vertex approx position) correct with $y = 16$ marked or stated</i>	M1A1	2
	(iii)	Line of symmetry $x = 3$ <i>Must have correct equation</i>		
			B1	1
(c)		Translation (and no additional transf'n) through $\begin{bmatrix} 3 \\ 7 \end{bmatrix}$ <i>Not shift, move, transformation, etc</i>		
			E1	
		<i>one part correct eg 7 units up</i>		
			M1	
		<i>all correct – if not vector – must say 3 units in positive x-direction etc</i>		
			A1	3

[10]

Q19.

(a) (i) $\sqrt{x} = x^{\frac{1}{2}}$
Accept $p = 0.5$

B1

1

(ii) $\int \sqrt{x} \, dx = \frac{x^{1.5}}{1.5} \{+c\}$
Index raised by 1

M1

Correct ft on p. Condone missing '+c'

A1ft

2

(iii) Area = $\int_1^4 \sqrt{x} \, dx$
Limits 1 and 4 PI

B1

..... = $\frac{4^{1.5}}{1.5} - \frac{1}{1.5}$
 $F(4) - F(1)$

M1

= $\frac{14}{3}$

Accept 4.66 or better

A1

3

(b) (i) $y = x^{\frac{1}{2}} \Rightarrow \frac{dy}{dx} = \frac{1}{2} x^{-\frac{1}{2}}$
Index $(p - 1)$ ft

M1

When $x = 4$, $y'(4) = 0.25$
Attempt to find $y'(4)$

M1

When $x = 4$, $y = 2$

B1

Equation of tangent: $y - 2 = \frac{1}{4}(x - 4)$
accept other forms

A1

4

- (ii) When $x = 0, y = 1$ $B(0, 1)$
Subs $x = 0$ and then $y = 0$ into equation of tangent. PI

M1

When $y = 0, x = -4$ $A(-4, 0)$
*Correct ft y_B and x_A
 (may be awarded as part of area calculation)*

A1ft

Area = $0.5(1)(4) = 2$
ft wrong sloping tangent and max of 1 further slip. Final answer must be +ve

A1ft

3

- (c) Translation
'Translation'/'translate(d)'

B1

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

*Accept equivalent in words provided linked to 'translation/move/shift'
 (Note: B0B1 is possible)*

B1

2

- (d) $h = 1$

PI

B1

Integral = $h/2 \{ \dots \}$
 $\{ \dots \} = f(1) + 2[f(2) + f(3)] + f(4)$
OE summing of areas of the three traps

M1

$$\{...\} = 0 + 2(1 + \sqrt{2}) + \sqrt{3}$$

Condone 1 numerical slip

A1

$$\text{Integral} = \frac{1}{2} \{2(1 + 1.414...) + 1.732\}$$

$$\text{Integral} = 0.5 \times 6.560... = 3.28 \text{ to 3sf}$$

Must be 3.28

A1CAO

4

[19]

Q20.

- (a) Stretch (parallel) to x -axis

B1

$$\text{Scale factor } \frac{1}{2}$$

B1

$$\text{Translate } \begin{pmatrix} 0 \\ 3 \end{pmatrix}$$

B1,B1

4

- (b) $A = \int e^{2x} + 3 \, dx$
(+ attempt to integrate)

M1

$$= \left[\frac{1}{2} e^{2x} + 3x \right]$$

(correct)

A1

$$\left(\frac{1}{2} e^8 + 12 \right) - \left(\frac{1}{2} e^4 + 6 \right)$$

Substitute 2,4 into their [

m1

$$\frac{1}{2}(e^8 - e^4) + 6$$

$$\left(\frac{1}{2}e^4(e^4 - 1) + 6 \right)$$

A1

4

- (c) $x_1 = 2, \quad y_1 = e^4 + 3 \quad (57.6)$
Attempt at y(2) or y(4)

M1

$x_2 = 4, \quad y_2 = e^8 + 3 \quad (2980)$
Both correct

A1

Area of $A + B =$
 $2(e^8 - e^4) + 2(e^8 + 3)$
Attempt to find correct area

M1

Area $B =$
 $4e^8 - 2e^4 + 6$
 $-\frac{1}{2}e^8 + \frac{1}{2}e^4 - 6$
 $= \frac{7}{2}e^8 - \frac{3}{2}e^4$

A1

4

[12]